Illuminating Darkness: Learning to Enhance Low-light Images In-the-Wild

S M A Sharif[1], Abdur Rehman[1], Zain Ul Abidin[2], Fayaz Ali Dharejo[3], Radu Timofte[3], Rizwan Ali Naqvi[2]
[1]Opt-AI Inc. [2]Sejong University [3]University of Würzburg

Code&Dataset

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Problem Formulation

SLLIE Dataset

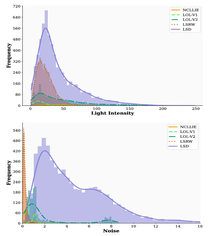
Limited real-world representation: Existing datasets [1-2] use synthetic low-light images (low ISO settings, artificially dimmed scenes) that fail to capture real-world phenomena like sensor noise, spatially varying light intensity, and color distortions from high-ISO amplification

Constrained distributions: Datasets exhibit limited dynamic range in noise and intensity, inadequately representing stochastic lighting conditions

Enhancement Method

RGB/Retinex methods: RGB models entangle semantics with noise (color instability, texture loss); Retinex methods [3] misestimate illumination (halo artifacts)

Non-RGB methods (e.g., HVI [4]): Separate luminance/chrominance but suffer hue discontinuities and chroma noise from limited cross-domain interaction



Noise and intensity distribution comparison between LSD and prior SLLIE datasets

Objective: Collect a large low-light enhancement dataset in uncontrolled real-world and a robust method that can handle challenging enhancement task

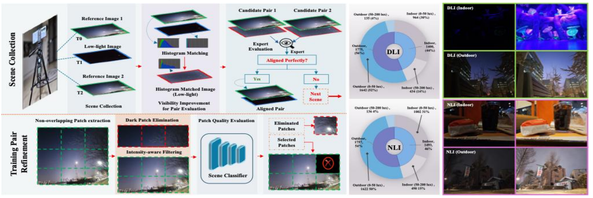
Method

LSD

Step 1: Multi-Frame Acquisition (The Capture)
- **"Sandwich" Cap.:** High-Low-High sequence yields aligned NLI (Noisy) and DLI (Denoised) pairs.
- *Why?* Guarantees temporal alignment while covering diverse noise patterns (Raw vs. ISP).

Step 2: Expert Verification (The Align)
- **Manual Review:** Experts inspect histogram-boosted pairs to detect micro-movements.
- *Why?* Discards ~1,500 scenes with focal shifts or motion blur that ruin training.

Step 3: Training Patch (local) Filtering (The Refine)
- **VGG Confidence Filter:** deep classifier to remove patches with low quality scores (C < 0.90).
- *Why?* Prevents the model from learning from blurry or under-exposed reference data.

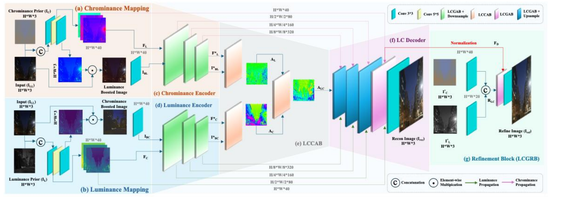


TFFormer

Step 1: LC Decomposition (The Split)
- Input is separated into Luminance (I_L) and Chrominance (I_C) (a-b).
- *Why?* Prevents noise in color channels from corrupting structural details.

Step 2: Dual-Path Encoding (The Fork)
- Two separate encoders process Structure (c) and Color (d) independently.
- *Why?* Allows the model to fix color casts without blurring edges.

Step 3: Fusion & Refinement (The Merge)
- **LCCAB:** Adaptively fuses the two streams using cross-attention (e).
- **LCGRB and LCCG Loss:** Ensure global consistency and removes remaining artifacts (g).



Generalization

Enhancing Unseen Images with LSD-TFFormer

Our model generalizes to **video streams** and revives **legacy content**
- **Video Adaptation:** Adapts seamlessly to video frames despite being trained only on static images.
- **Legacy Restoration:** Revitalizes low-quality photos from older sensors (e.g., Samsung Note 8).
- **Artifact Removal:** Eliminates compression noise from decade-old social media archives.

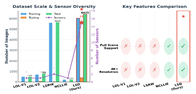
Device	Low-light ↑	LSD-TFFormer ↑
DSLR	0.0952	0.9048
Video (Frames)	0.3095	0.6905
iPhone	0.1905	0.8095
Android	0.2063	0.7937
Social Media	0.0714	0.9286
Average	0.1746	0.8254

82% of participants preferred the LSD-Tformer enhanced images over their low-light counterparts.

Illuminating Darkness

Low-light Smartphone Dataset (LSD)
- **Scale:** 6,425 high-res pairs from 15 smartphone sensors.
- **Diversity:** Diverse lighting (0.1 - 200 lux).
- **Generalization:** 2,117 unpaired images for "in-the-wild" testing.

LSD vs SLLIE Dataset

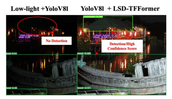


Tuning-fork-shaped Transformer (TFFormer)
- **Core Idea:** A "Tuning Fork" Transformer that decouples **Luminance** (Structure) and **Chrominance** (Color).
- **LCCAB:** Cross-attention block to fuse features adaptively.
- **LCGRB, and LCCG:** for perceptually consistent enhancement.

Unseen SLLIE

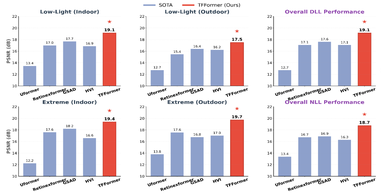
Key Impact
- **SOTA:** Achieves **+2.45 dB PSNR** over previous best methods.
- **Generalization:** Generalized across unseen scenes.
- **Downstream:** Boosts Object Detection (ExDark) by **+6.80 mAP**.

Vision Task

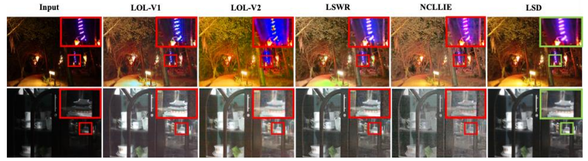


Experiments & Results

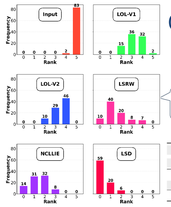
Qualitative comparison of existing SLLIE methods on the LSD dataset. TFFormer produces cleaner, more plausible images, outperforming existing methods.



TFFormer consistently outperforms the strong baseline RetinexFormer by **+2.04 dB** and the recent HVI by **+2.26 dB** in PSNR, with similar gains in SSIM and LPIPS.



Qualitative comparison of real-world SLLIE leveraging TFFormer trained on different datasets.



Cross-dataset Evaluation

Downstream vision task (object detection) imporvement

User preference study on SLLIE datasets

Ablation (LSD)

Scene Type	DLI			NLI			LSD (Average)		
	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
Raw Patch	15.47	0.6330	0.3107	15.80	0.6293	0.3155	15.64	0.6312	0.3131
Full-scene	17.52	0.6396	0.2770	17.09	0.6095	0.3239	17.31	0.6246	0.3005
Filtered Patch	19.13	0.7091	0.2609	18.74	0.6995	0.2661	18.93	0.7043	0.2635

Conclusion

LSD: 6,425 high-res pairs from 15 sensors → ~180K training patches + 400 test pairs + 2,117 unpaired images

TFFormer: Luminance-chrominance decoupling via **LCCAB/LCGRB/LCPL** modules for superior color restoration and denoising

Generalization: Strong generalization across different **devices and sources**

Limitation: Observed artefacts on **extreme noise** (under 1 lux)

Future Scope: Planned to extend for **joint** detection and enhancement

Reference

[1] Wei et al., Deep Retinex Decomposition for Low-Light Enhancement, arXiv 2018

[2] Hai et al., R2RNet: Real-Low to Real-Normal Network, JVCIR 2023

[3] Cai et al., Retinexformer: One-Stage Retinex Transformer, ICCV 2023

[4] Yan et al., HVI: New Color Space for Low-Light Enhancement, CVPR 2025